

QuickPlot

SWE 4404: Software Project Lab II

Group 09

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Introduction

QuickPlot



A web-based graphing platform that renders mathematical equations interactively.



Plot graph for Cartesian and polar coordinate systems both in 2D & 3D spaces.



Workspaces can be saved and shared for distributions.



Platform for visual, intuitive math learning

Motivation

Why QuickPlot?



Learning Gap

Students struggle to visualize math equations instantly



Context Switching

Learners reading notes had to leave the page for a graph function



Collaboration

Sharing links or data formats ease the collaboration



Skill Building

Production-grade graphing builds real-world engineering skills

Features

CORE



Equation Grapher

- ☐ Rendering to Graph
- ☐ Adding Multiple functions



Parameter Sliders

- ☐ Adding Sliders



Zoom & Pan

- ☐ Zooming Canvas
- ☐ Pan the Graph



Implicit Equations

- ☐ Rendering by Interpolation



Polar Graphing

- ☐ Adding Polar mode



Save & Share

- ☐ Export
- ☐ Sharing links

Features

Extended



Themes

- ☐ Dark & Light Mode
- ☐ Featured Themes



Authentication

- ☐ Register / Sign up
- ☐ Login
- ☐ Encryption



User Responsive

- ☐ Adaptive Layout



Extension

- ☐ Browser Extension



Data Visualization

- ☐ Import dataset
- ☐ Rendering by Interpolation

Application

Who uses QuickPlot?



Students

- ☐ Graph equations instantly from textbooks
- ☐ No app switching or loading waits



Educators

- ☐ Live demos with sliders
- ☐ Visualize concepts instantly



Researchers

- ☐ Quick model visualization
- ☐ Easy result sharing

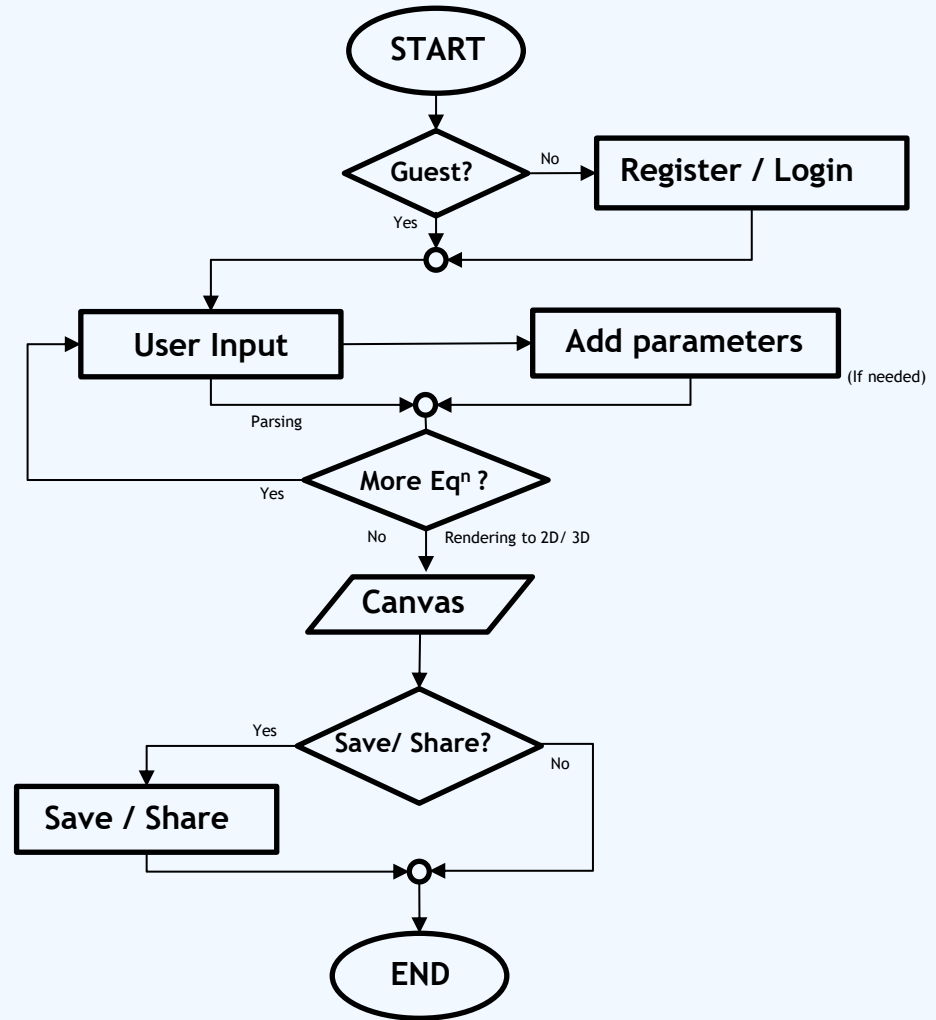


Developers

- ☐ Test functions visually
- ☐ Explore edge cases fast

Flowchart

QuickPlot



Tools & Technology

Core
Language



TypeScript



JavaScript

Frontend



React



Tailwind CSS

Rendering
Engine



HTML5 Canvas



Three.js



D3-scale

Tools & Technology

Backend



Supabase



Vercel

Database



PostgreSQL



Supabase

Extension



Chrome MV3

DevOps



Git



GitHub



VSCode

Project Timeline





Thank You

Looking for Feedbacks

QuickPlot